

# Juhyun Jung

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|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| RESEARCH<br>INTERESTS  | Robotics, Manipulation, Motion Planning, Underwater Robotics, Bio-Inspired Robotics                                                                                                                                                                                                                                                                                                                                                                           |                      |
| EDUCATION              | <b>Stanford University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | Stanford, CA         |
|                        | M.S. in Mechanical Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                | Sept. 2025–Present   |
|                        | • GPA: 4.06/4.00; Advisor: Prof. Mark Cutkosky and Prof. Shuran Song                                                                                                                                                                                                                                                                                                                                                                                          |                      |
|                        | <b>Illinois Institute of Technology</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | Chicago, IL          |
|                        | B.S. in Mechanical Engineering, Minor in Artificial Intelligence                                                                                                                                                                                                                                                                                                                                                                                              | Aug. 2022–Dec. 2024  |
|                        | • GPA: 4.00/4.00, <i>Dean's List</i>                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |
|                        | <b>Ajou University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        | Suwon, South Korea   |
|                        | B.S. in Mechanical Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                | Mar. 2018–Dec. 2024  |
|                        | • GPA: 4.47/4.50, <i>ranked 1st in class (1/170)</i> , <i>Dean's List</i>                                                                                                                                                                                                                                                                                                                                                                                     |                      |
| RESEARCH<br>EXPERIENCE | <b>Biomimetics and Dextrous Manipulation Lab (BDML)</b> , Stanford University                                                                                                                                                                                                                                                                                                                                                                                 |                      |
|                        | Graduate Research Assistant · Advisor: Prof. Mark Cutkosky · Stanford, CA                                                                                                                                                                                                                                                                                                                                                                                     | Sept. 2025–Present   |
|                        | <b>UMI-Underwater 2.0: Robust Manipulation (Project Lead, Ongoing)</b>                                                                                                                                                                                                                                                                                                                                                                                        | Apr. 2026–Present    |
|                        | Co-advised by Prof. Shuran Song                                                                                                                                                                                                                                                                                                                                                                                                                               |                      |
|                        | <ul style="list-style-type: none"><li>Leading an ongoing project to develop robust control policies that enable an underwater manipulator to achieve stable grasping under hydrodynamic disturbances.</li><li>Designing an IMU–vision Extended Kalman Filter (EKF) for state estimation.</li><li>Developing low-level controllers using reinforcement learning (RL) and model predictive control.</li></ul>                                                   |                      |
|                        | <i>Fiber Bragg Grating Whiskers for Bioinspired Hydrodynamic Perception on Underwater Robots</i>                                                                                                                                                                                                                                                                                                                                                              |                      |
|                        | Under review, <i>IEEE Transactions on Robotics (T-RO)</i> · <a href="https://arxiv.org/abs/2608.24724">arXiv:2608.24724</a>                                                                                                                                                                                                                                                                                                                                   | Sept. 2025–Apr. 2026 |
|                        | <ul style="list-style-type: none"><li>Developed a novel tactile sensing modality inspired by animal whiskers, integrating fiber Bragg grating (FBG) optical strain sensors with compliant beams for underwater flow detection.</li><li>Improved the signal-to-noise ratio by over 8× through experimental design and signal processing.</li><li>Designed and built an embedded testbed with stepper-motor actuation and real-time data acquisition.</li></ul> |                      |
|                        | <b>BUDDY: Bimanual Underwater Demonstration-Driven Policy (Ongoing)</b>                                                                                                                                                                                                                                                                                                                                                                                       | 2026–Present         |
|                        | Poster at IROS 2026 Workshop, Pittsburgh, PA                                                                                                                                                                                                                                                                                                                                                                                                                  |                      |
|                        | <ul style="list-style-type: none"><li>Designing the data collection pipeline for synchronized bimanual demonstrations with an instrumented gripper to identify the minimum dexterity and sensing for underwater manipulation.</li><li>Designing a diffusion policy architecture that fuses fingertip tactile sensing with vision for bimanual cable connection on two Franka arms, toward in-air to underwater transfer.</li></ul>                            |                      |
|                        | <b>Computer Vision and Multimedia Laboratory (CVM Lab)</b> , Illinois Institute of Technology                                                                                                                                                                                                                                                                                                                                                                 |                      |
|                        | Undergraduate Research Assistant · Advisor: Prof. Yan Yan · Chicago, IL                                                                                                                                                                                                                                                                                                                                                                                       | June 2023–Dec. 2024  |
|                        | <ul style="list-style-type: none"><li>Developed an LSTM-based algorithm to classify mouse behavior from pose datasets.</li><li>Built a Vision Transformer (ViT) classifier over full mouse video datasets, improving behavior prediction accuracy.</li><li>Supported PhD students in evaluating multimodal models, including concatenation fusion and gated information fusion, for behavior classification.</li></ul>                                        |                      |
|                        | <b>Multiscale Bio-Inspired Technology Laboratory (MOST Lab)</b> , Ajou University                                                                                                                                                                                                                                                                                                                                                                             |                      |
|                        | Undergraduate Researcher · Advisor: Prof. Je-Sung Koh · Suwon, South Korea                                                                                                                                                                                                                                                                                                                                                                                    | Jan. 2022–June 2022  |
|                        | <ul style="list-style-type: none"><li>Designed and built a comprehensive experimental setup for dielectric elastomer actuator (DEA) research.</li></ul>                                                                                                                                                                                                                                                                                                       |                      |

- Characterized DEA performance across frequency, electrode area, elastomer composition and thickness, voltage, and pre-strain, demonstrating a maximum strain of 12.3%.

#### PREPRINTS

1. H. Li, T. Tu, S. Yao, Z. Chang, **J. Jung**, X. Xie, L. Y. Chung, T.-A. Ren, G. Chen, and M. Cutkosky, "Fiber Bragg Grating Whiskers for Bioinspired Hydrodynamic Perception on Underwater Robots", under review at *IEEE Transactions on Robotics (T-RO)*, 2026. [arXiv:2608.24724](https://arxiv.org/abs/2608.24724).

#### PROJECTS

**Understanding the Varied Economic Impact of Tax Increment Financing (TIF)** Jan.–May 2024  
Advisor: Prof. Robert Ellis · Consulting expert: Tom Tresser (Co-Founder, CivicLab)

- Analyzed the effect of TIF districts on median income in Chicago, finding significant gains for the White population, concentrated in White-dominant, high-income districts.

**Autonomous Mobility Competition — Vision-Based Lane Following** Sept.–Dec. 2021

- Designed a vision-based lane-following algorithm combining image processing and PID control to optimize real-time model-car performance; awarded 2nd place by HL Mando Corporation.

**System Innovation Competition** · Advisor: Prof. Peom Park Mar.–May 2021

- Analyzed VR limitations such as cybersickness and designed solutions including accelerometer integration and sound effects, supported by market surveys; awarded 3rd place by KOSSE.

#### PRESENTATIONS

1. S. Choi, V. Patel, L. Klostermair, **J. Jung**, H. Watkins, O. Khatib, and M. Cutkosky, "BUDDY: Bimanual Underwater Demonstration-Driven Policy", *IROS 2026 Workshop*, Pittsburgh, PA, 2026. (Poster)
2. J. Jung, "Autonomous control using automated image analysis", *Fall 2024 Armour R&D Expo*, Chicago, IL, Nov. 2024. (Poster)
3. J. Jung, "Autonomous control using automated image analysis", *Summer 2024 Armour R&D Research Immersion Exposition*, Chicago, IL, July 2024. (Poster)
4. J. Jung, "Understanding the varied economic impact of Tax Increment Financing (TIF)", *Ed Kaplan Family Institute Innovation Day*, Chicago, IL, Apr. 2024. (Poster)
5. J. Jung, "Assessment of cybersickness in VR social media services", *The Korean Society of Systems Engineering 2021 Conference*, Jeju, South Korea, June 2021. (Poster)

#### HONORS & AWARDS

|                                                                                         |                  |
|-----------------------------------------------------------------------------------------|------------------|
| <b>Dean's List</b> , Illinois Institute of Technology                                   | 2022–2024        |
| <b>Dean's List</b> , Ajou University                                                    | 2018, 2021, 2022 |
| <b>Academic Excellence Scholarship</b> , Ajou University                                | 2018, 2021, 2022 |
| <b>Department Chair's Award for Academic Excellence</b> , Ajou University               | 2019, 2022       |
| <b>2nd Place, Autonomous Mobility Competition</b> , HL Mando Corporation                | 2021             |
| <b>3rd Place, System Innovation Competition</b> , Korean Society of Systems Engineering | 2021             |
| <b>Soldier of Valor Award</b> , Republic of Korea Army                                  | 2020             |
| <b>Active Learner Self-Development Scholarship</b> , Ajou University                    | 2018             |

#### SKILLS & COURSEWORK

**Programming & Tools:** Python, C/C++, MATLAB, Julia, ROS 2, Linux (Ubuntu 22.04)  
**Robotics:** Robot Autonomy I (CS 237A), Collaborative Robotics (ME 326), Introduction to Robotics (CS 223A), Optimal and Learning-Based Control (AA 203)  
**Machine Learning:** Deep Learning (CS 230), Decision Making under Uncertainty (CS 238), Deep Reinforcement Learning (CS 224R)

#### ACTIVITIES

- Balanced Man Scholarship Chair, Sigma Phi Epsilon Fraternity (Illinois Beta)
- Member, Tau Beta Pi Engineering Honor Society
- Member, Climbing Club